

HARDWARE ARCHITECTURE SPECIFICATION

AES-256 Cryptographic IP Core with AXI4-Lite Interface

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1. Introduction

1.1 Purpose

This Hardware Architecture Specification (HAS) describes the microarchitecture, interface protocol, register map, timing requirements, and programming model of the **AES-256 Cryptographic IP Core** produced by Antigravity Silicon Inc. (ASI). The document serves as the authoritative reference for SoC integration engineers, firmware developers, and verification engineers working with this IP.

1.2 Scope

This document covers:

- Architectural overview and block diagram description.
- Top-level port list and AXI4-Lite interface protocol.
- Complete 32-bit register map with field-level descriptions.
- Functional description of all six modes of operation (ECB, CBC, CTR, CFB, OFB, GCM).
- Key expansion and datapath microarchitecture.
- Timing diagrams for initialization, single-block, and streaming operations.
- Power, area, and performance characteristics.
- Integration guidelines, clock-domain considerations, and reset sequences.
- Firmware programming model with annotated pseudocode examples.

1.3 Applicable Documents

Document	Title	Revision
FIPS PUB 197	Advanced Encryption Standard (AES)	Nov 2001
NIST SP 800-38A	Recommendation for Block Cipher Modes of Operation	2001 Ed.
NIST SP 800-38D	Recommendation for Block Cipher Modes: GCM and GMAC	Nov 2007
ARM IHI0022E	AMBA AXI and ACE Protocol Specification AXI3, AXI4	Rev E
ASI-INT-0001	Antigravity Silicon SoC Integration Guidelines	v2.3

1.4 Terminology and Abbreviations

Term / Acronym	Definition
AAD	Additional Authenticated Data (GCM mode).
AES	Advanced Encryption Standard (FIPS-197).
APB	Advanced Peripheral Bus.
AXI	Advanced eXtensible Interface (AMBA specification).
CBC	Cipher Block Chaining mode.
CFB	Cipher Feedback mode.

CTR	Counter mode.
DUT	Device Under Test.
ECB	Electronic Codebook mode.
FIFO	First-In First-Out buffer.
GCM	Galois/Counter Mode — authenticated encryption with associated data.
GMAC	Galois Message Authentication Code.
HAS	Hardware Architecture Specification (this document).
IV	Initialization Vector (also referred to as Nonce in CTR/GCM modes).
IP	Intellectual Property (reusable hardware block).
IRQ	Interrupt Request.
LSB	Least Significant Bit.
MSB	Most Significant Bit.
OFB	Output Feedback mode.
RTL	Register-Transfer Level (hardware description).
SoC	System-on-Chip.
TAG	128-bit GCM authentication tag.
UVM	Universal Verification Methodology.

2. Overview

2.1 Functional Description

The ASI AES-256 IP Core is a fully synchronous, synthesizable RTL implementation of the Advanced Encryption Standard operating exclusively in 256-bit key mode. It is designed for integration into standard AMBA-based SoC fabrics through its 32-bit AXI4-Lite slave port. The IP exposes all control, status, and data registers through this bus, enabling direct CPU-driven or DMA-assisted cryptographic operations.

The core supports all six standard modes of operation as defined by NIST SP 800-38A and SP 800-38D: ECB, CBC, CTR, CFB, OFB, and GCM. GCM mode additionally provides 128-bit authenticated tag generation and verification compatible with GMAC.

2.2 Key Features

- **FIPS-197 Compliant:** Fully validated AES-256 encrypt/decrypt datapath.
- **Six Modes of Operation:** ECB, CBC, CTR, CFB, OFB, and GCM with AAD support.
- **AXI4-Lite Interface:** 32-bit slave interface with 256-byte address space.
- **Interrupt Driven:** Configurable IRQ on operation complete and authentication tag ready.
- **High Throughput:** 14-cycle latency per 128-bit block in non-feedback modes.
- **Back-to-Back Streaming:** Hardware chaining state eliminates re-initialization overhead between consecutive blocks.
- **Low Area:** Shared S-Box architecture minimizes silicon footprint; single-datapath, non-pipelined design.
- **Configurable Clock:** Operates from 10 MHz to 400 MHz depending on process/VT corner.
- **Active-Low Reset:** Asynchronous external reset with synchronous internal de-assertion.

2.3 Top-Level Block Diagram Description

The IP core comprises four major subsystems interconnected through a shared internal bus fabric:

- **AXI4-Lite Interface Controller (AIC):** Decodes AXI4-Lite transactions, manages the register file, and drives the interrupt output. It also provides a buffered read-back register set for DOUT and TAG to allow stable reads even while the core begins processing a subsequent block.
- **Key Expansion Unit (KEU):** Accepts the 256-bit user key from the KEY registers and computes all 15 round key words (240 bytes total) using the AES key schedule algorithm. Key expansion completes in approximately 60 clock cycles and the result is stored in an on-chip round-key RAM accessible to the datapath.
- **AES Encrypt/Decrypt Datapath (ADP):** An iterative 128-bit datapath executing one round of AES per clock cycle. The datapath shares four 256-entry S-Box ROMs between encrypt and decrypt paths. A 14-cycle pipeline controller sequences SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
- **Mode Wrapper Logic (MWL):** Implements the XOR chaining, counter increment, Galois field multiplication, and GHASH accumulation logic required by each supported mode of operation. The MWL sits between the AIC and ADP, intercepting data before and after processing to apply mode-specific transforms.

2.4 Deliverables

Deliverable	Format	Notes
-------------	--------	-------

Synthesizable RTL Source	SystemVerilog (.sv)	Encrypted via IEEE 1735.
UVM Verification Environment	SystemVerilog + SV (.sv)	Includes directed and random tests.
Synthesis Scripts	Synopsys Design Compiler TCL	TSMC 28nm HPC+ target.
Place & Route Netlist (Optional)	Verilog Netlist (.v)	Post-PnR, back-annotated.
FIPS Known-Answer Test Vectors	Plain text (.txt / .csv)	All modes, all NIST vectors.
This HAS Document	.docx / .pdf	Revision controlled.

3. Interface Description

3.1 Port List

All ports are synchronous to `i_aes_aclk` unless explicitly noted. The reset `i_aes_aresetn` is asynchronous-assert, synchronous-deassert (reset synchronizer included inside the IP).

3.1.1 Global Signals

Signal	Width	Dir.	Description
<code>i_aes_aclk</code>	1	In	Primary clock. All internal logic is triggered on the rising edge.
<code>i_aes_aresetn</code>	1	In	Active-low asynchronous reset. Clears all internal state registers, mode logic, and interrupt flags.
<code>o_aes_irq</code>	1	Out	Level-high interrupt. Asserted based on <code>IRQ_EN</code> mask and <code>IRQ_STAT</code> events. Must be cleared by writing 1 to <code>IRQ_STAT</code> .

3.1.2 AXI4-Lite Write Address Channel (AW)

Signal	Width	Dir.	Description
<code>i_aes_awvalid</code>	1	In	Master asserts to indicate write address is valid.
<code>o_aes_awready</code>	1	Out	Slave asserts when ready to accept the write address.
<code>i_aes_awaddr</code>	8	In	Write address, byte-addressable, 8-bit to cover 256-byte register space.
<code>i_aes_awprot</code>	3	In	Protection type. Informational only; not used by this core.

3.1.3 AXI4-Lite Write Data Channel (W)

Signal	Width	Dir.	Description
<code>i_aes_wvalid</code>	1	In	Master asserts to indicate write data is valid.
<code>o_aes_wready</code>	1	Out	Slave asserts when ready to accept write data.
<code>i_aes_wdata</code>	32	In	Write data bus. Little-endian; least significant byte at lowest address.
<code>i_aes_wstrb</code>	4	In	Write byte strobes. Each bit corresponds to one byte of <code>wdata</code> . All 4 bits must be asserted for register writes (partial strobes are not supported).

3.1.4 AXI4-Lite Write Response Channel (B)

Signal	Width	Dir.	Description
<code>o_aes_bvalid</code>	1	Out	Asserted to indicate write response is valid.
<code>i_aes_bready</code>	1	In	Master asserts when ready to accept the write response.
<code>o_aes_bresp</code>	2	Out	Write response code. <code>2'b00</code> = OKAY. <code>2'b10</code> = SLVERR (write to read-only register).

3.1.5 AXI4-Lite Read Address Channel (AR)

Signal	Width	Dir.	Description
i_aes_arvalid	1	In	Master asserts to indicate read address is valid.
o_aes_arready	1	Out	Slave asserts when ready to accept the read address.
i_aes_araddr	8	In	Read address, byte-addressable, 8-bit.
i_aes_arprot	3	In	Protection type. Informational only.

3.1.6 AXI4-Lite Read Data Channel (R)

Signal	Width	Dir.	Description
o_aes_rvalid	1	Out	Asserted to indicate read data is valid.
i_aes_rready	1	In	Master asserts when ready to accept read data.
o_aes_rdata	32	Out	Read data. Reflects the register value at the sampled address.
o_aes_rresp	2	Out	Read response. 2'b00 = OKAY. 2'b10 = SLVERR (read from write-only register returns 0 with SLVERR).

NOTE

This IP does not support outstanding (out-of-order) AXI4-Lite transactions. The master must wait for BVALID/RVALID before initiating a new transaction to the same address channel.

4. Register Map

The IP occupies a 256-byte (0x00–0xFF) address space. All registers are 32-bit wide, 4-byte aligned. Unimplemented bits read as 0 and are ignored on write unless noted. Accesses to undefined offsets return SLVERR on the AXI response channel.

CAUTION

Writing to KEY registers while BUSY (STATUS[0] = 1) will corrupt the key expansion state. Always wait for BUSY to deassert before loading a new key.

4.1 Register Summary

Offset	Name	Access	Description
0x000	CTRL	R/W	Control: START, ENC_DEC, MODE[2:0], INIT_IV.
0x004	STATUS	R	Status: BUSY, DONE, KEY_READY, AUTH_FAIL, GCM_PHASE.
0x008	IRQ_EN	R/W	Interrupt Enable mask register.
0x00C	IRQ_STAT	R/W1C	Interrupt Status — Bit 0: DOUT_VLD, Bit 1: TAG_VLD. Write 1 to clear.
0x010	KEY[0]	W	Key word 0 — bits [31:0] of 256-bit key (MSB first).
0x014	KEY[1]	W	Key word 1 — bits [63:32].
0x018	KEY[2]	W	Key word 2 — bits [95:64].
0x01C	KEY[3]	W	Key word 3 — bits [127:96].
0x020	KEY[4]	W	Key word 4 — bits [159:128].
0x024	KEY[5]	W	Key word 5 — bits [191:160].
0x028	KEY[6]	W	Key word 6 — bits [223:192].
0x02C	KEY[7]	W	Key word 7 — bits [255:224] (LSB of key).
0x030	IV[0]	W	IV/Nonce word 0 — bits [31:0] (MSB first).
0x034	IV[1]	W	IV/Nonce word 1 — bits [63:32].
0x038	IV[2]	W	IV/Nonce word 2 — bits [95:64].
0x03C	IV[3]	W	IV/Nonce word 3 — bits [127:96].
0x040	DIN[0]	W	Data Input word 0 — bits [31:0] (MSB first).
0x044	DIN[1]	W	Data Input word 1.
0x048	DIN[2]	W	Data Input word 2.
0x04C	DIN[3]	W	Data Input word 3. Write triggers processing.
0x050	DOUT[0]	R	Data Output word 0 — bits [31:0].
0x054	DOUT[1]	R	Data Output word 1.
0x058	DOUT[2]	R	Data Output word 2.
0x05C	DOUT[3]	R	Data Output word 3.
0x060	AAD[0]	W	Additional Auth. Data word 0 (GCM only).
0x064	AAD[1]	W	Additional Auth. Data word 1.

0x068	AAD[2]	W	Additional Auth. Data word 2.
0x06C	AAD[3]	W	Additional Auth. Data word 3. Write triggers GHASH update.
0x070	TAG[0]	R	Authentication Tag word 0 (GCM only).
0x074	TAG[1]	R	Authentication Tag word 1.
0x078	TAG[2]	R	Authentication Tag word 2.
0x07C	TAG[3]	R	Authentication Tag word 3.
0x080	DIN_CTRL	R/W	Data control flags: Bit 0 = DIN_LAST, Bit 1 = AAD_LAST.
0x084	GCM_LEN_A	W	GCM AAD bit-length (lower 32 bits of 64-bit len_A).
0x088	GCM_LEN_A_HI	W	GCM AAD bit-length (upper 32 bits of 64-bit len_A).
0x08C	GCM_LEN_C	W	GCM ciphertext bit-length (lower 32 bits).
0x090	GCM_LEN_C_HI	W	GCM ciphertext bit-length (upper 32 bits).
0x0FC	VERSION	R	IP Version: [31:16]=Major, [15:8]=Minor, [7:0]=Patch.

4.2 Register Descriptions

4.2.1 Control Register (CTRL — 0x000)

Reset value: 0x00000000

Bits	Field	Access	Description
[0]	START	W1S	Write 1 to commence operation. Self-clears after one clock cycle. Ignored if BUSY is asserted.
[1]	ENC_DEC	R/W	1 = Encrypt; 0 = Decrypt. Must be set before asserting START. Changing this field while BUSY is undefined behavior.
[4:2]	MODE	R/W	Mode of Operation: 3'b000 = ECB, 3'b001 = CBC, 3'b010 = CTR, 3'b011 = CFB, 3'b100 = OFB, 3'b101 = GCM. Values 3'b110–3'b111 are reserved.
[5]	INIT_IV	R/W	1 = Load IV registers into the chaining state at the next START. Must be set for the first block of a new message in CBC/CTR/CFB/OFB/GCM. Should be cleared for back-to-back streaming of subsequent blocks.
[31:6]	—	R	Reserved. Reads as 0.

4.2.2 Status Register (STATUS — 0x004)

Reset value: 0x00000000. Read-only; writes are ignored.

Bits	Field	Description
[0]	BUSY	1 = Core is actively processing (AES rounds, key expansion, or GHASH update). Software must not write to any data or key register while BUSY is set.
[1]	DONE	1 = The current block operation has completed and output data is valid in DOUT. Cleared when the next START is written or IRQ_STAT[0] is cleared.
[2]	KEY_READY	1 = Key expansion is complete and round keys are loaded. Cleared when new data is written to any KEY register.
[3]	AUTH_FAIL	GCM decrypt only. 1 = The computed TAG does not match the expected TAG loaded via TAG registers. This flag is sticky until the next START.
[6:4]	GCM_PHASE	GCM mode only. 3'b000 = Idle, 3'b001 = AAD processing, 3'b010 = Ciphertext processing, 3'b011 = Tag finalization.
[31:7]	—	Reserved.

4.2.3 Interrupt Enable Register (IRQ_EN — 0x008)

Reset value: 0x00000000. All interrupt sources are disabled after reset.

Bits	Field	Description
[0]	DOUT_VLD_EN	1 = Enable interrupt when a data output block is ready (IRQ_STAT[0] sets and o_aes_irq asserts).
[1]	TAG_VLD_EN	1 = Enable interrupt when the GCM authentication tag is finalized (IRQ_STAT[1] sets and o_aes_irq asserts).
[31:2]	—	Reserved.

4.2.4 Interrupt Status Register (IRQ_STAT — 0x00C)

Reset value: 0x00000000. Write-1-to-clear (W1C). Writing 0 to a bit has no effect.

Bits	Field	Description
[0]	DOUT_VLD	Set when DOUT is populated with a valid output block. Write 1 to clear and de-assert o_aes_irq (if TAG_VLD is also 0). Clearing this bit also signals to the core that the output has been consumed, enabling streaming.
[1]	TAG_VLD	Set when the GCM TAG registers contain the finalized authentication tag. Write 1 to clear.
[31:2]	—	Reserved.

4.2.5 Data Input Control Register (DIN_CTRL — 0x080)

Bits	Field	Description
[0]	DIN_LAST	Indicates the current DIN block is the last plaintext/ciphertext block of the message. Required for GCM to trigger tag finalization. Self-clears after the block is processed.
[1]	AAD_LAST	GCM only. Indicates the current AAD block is the last AAD block. Must be set before writing the final AAD[3]. Self-clears after GHASH update.
[31:2]	—	Reserved.

5. Theory of Operation

5.1 Key Expansion Unit (KEU)

The KEU implements the AES-256 key schedule as defined in FIPS-197 Section 5.2. Upon detecting a write to `KEY[7]` (offset 0x02C), the KEU automatically initiates the key expansion process. The 256-bit user key is expanded into 15 round keys (each 128 bits), totaling 1920 bits stored in an internal dual-port SRAM of 60×32 -bit words.

The expansion engine processes one round constant (Rcon) per two clock cycles. Total key expansion requires approximately 60 clock cycles. During expansion, `STATUS[0]` (`BUSY`) is asserted and `STATUS[2]` (`KEY_READY`) is deasserted. Software must poll `KEY_READY` or rely on a deferred start mechanism before commencing data processing.

NOTE

Key registers are write-only. A read to any `KEY[n]` offset will return 0x00000000 with an AXI SLVERR response to prevent key material leakage through the bus.

5.2 AES Encrypt/Decrypt Datapath (ADP)

The ADP implements an iterative, non-pipelined AES round function. The datapath processes one AES round per clock cycle. For AES-256, 14 rounds are required, yielding a block processing latency of exactly 14 clock cycles from the write to `DIN[3]` until `STATUS[1]` (`DONE`) is asserted.

The datapath components are:

- **SubBytes:** Implemented using four parallel 256×8 -bit synchronous ROMs (S-Box lookup tables), shared between encrypt and decrypt paths using a mux. This approach reduces gate count at the cost of one additional cycle for mux settling on direction change.
- **ShiftRows:** Implemented as hardwired byte-lane rearrangement with zero logic cost.
- **MixColumns:** Implemented using $GF(2^8)$ polynomial arithmetic with the irreducible polynomial $x^8+x^4+x^3+x+1$ (0x11B). An `InvMixColumns` variant shares the multiplication units via polarity control.
- **AddRoundKey:** 128-bit XOR with the current round key fetched from the round-key SRAM.

5.3 Modes of Operation

5.3.1 Electronic Codebook (ECB)

ECB mode applies the AES-256 cipher directly to each 128-bit input block without chaining. Each block is independently encrypted or decrypted. The IV registers are not used in ECB mode. ECB provides no semantic security for messages longer than one block and is included for completeness and backward compatibility only.

WARNING

ECB mode is deterministic and should NOT be used for encrypting data where patterns must be concealed. Use CBC, CTR, or GCM for general-purpose confidentiality.

5.3.2 Cipher Block Chaining (CBC)

In CBC encryption, each plaintext block is XORed with the previous ciphertext block before AES encryption. The first block uses the IV as the chaining value. The CBC chaining register is maintained internally and automatically updated between blocks when `INIT_IV` is cleared in `CTRL`.

In CBC decryption, each ciphertext block is first decrypted by AES and then XORed with the previous ciphertext block. The core stores the previous ciphertext block internally for use in `InvShiftRows` operations.

5.3.3 Counter (CTR)

CTR mode converts AES into a stream cipher by encrypting successive counter values and XORing the output with the plaintext. The 128-bit counter is initialized from the IV registers. The counter increment function adds 1 to the least significant 32 bits of the counter (little-endian increment), with carry propagation disabled beyond bit 31 (i.e., the counter wraps within its lower word).

CTR mode is the same for both encrypt and decrypt (ENC_DEC bit is ignored by the MWL; the ADP always performs encryption). This is configured internally.

5.3.4 Cipher Feedback (CFB) and Output Feedback (OFB)

CFB mode uses the previous ciphertext (or IV) as feedback into the AES cipher to generate a keystream, which is XORed with plaintext. OFB mode feeds back the AES output directly (not the ciphertext), producing a keystream independent of the data. Both modes are supported in full 128-bit segment mode.

5.3.5 Galois/Counter Mode (GCM)

GCM is an authenticated encryption mode combining CTR mode encryption with GHASH-based authentication. The core implements the full GCM specification per NIST SP 800-38D including:

- GHASH computation over AAD and ciphertext using $GF(2^{128})$ Galois field multiplication.
- A dedicated GHASH accumulator register (H) derived from encrypting the all-zero block with the current key.
- Support for AAD-only messages (GMAC) by setting DIN_LAST without writing any DIN data.
- Lengths field (len_A || len_C) appended to GHASH per the GCM specification using GCM_LEN registers.
- 128-bit TAG output available in TAG[0:3] after the final block is processed.

GCM decryption verifies the authentication tag by comparing the computed tag against the expected tag. The AUTH_FAIL status bit is set if they do not match. Software is responsible for discarding output data if AUTH_FAIL is set.

5.4 Interrupt Architecture

The IRQ architecture uses a two-register model: IRQ_EN (mask) and IRQ_STAT (event flags). The output `o_aes_irq` is the bitwise AND of the two registers' corresponding bits OR-reduced to a single signal. The interrupt is level-sensitive and remains asserted until all set bits in IRQ_STAT are cleared by the software.

Two interrupt sources are provided:

- `DOUT_VLD` (IRQ_STAT[0]): Asserted when a 128-bit output block is ready in DOUT registers.
- `TAG_VLD` (IRQ_STAT[1]): Asserted when the GCM authentication tag is finalized in TAG registers.

NOTE

It is possible for both `DOUT_VLD` and `TAG_VLD` to be set simultaneously in GCM mode for the last plaintext block. The ISR must check and clear both bits.

6. Programming Sequence

6.1 Reset and Initialization

After power-on or assertion of `i_aes_aresetn`, the following conditions are guaranteed:

- All registers are cleared to their reset values.
- `BUSY`, `DONE`, `KEY_READY`, `AUTH_FAIL`, and all IRQ flags are deasserted.
- The `o_aes_irq` output is deasserted.
- The chaining state (CBC IV, CTR counter, GCM H/GHASH accumulator) is cleared.

6.2 Key Loading

Key loading must be performed prior to any encryption or decryption operation. A new key may be loaded at any time as long as `BUSY` is deasserted.

1. **Step 1:** Write `KEY[0]` through `KEY[6]` (offsets 0x010–0x028) with key words 0–6.
2. **Step 2:** Write `KEY[7]` (offset 0x02C) with key word 7. This write triggers key expansion automatically.
3. **Step 3:** Poll `STATUS[2]` (`KEY_READY`) until it reads 1, or wait for a firmware-defined delay of at least 64 clock cycles.

```
// Example: Load 256-bit key (C pseudocode)
void aes_load_key(volatile uint32_t *base, const uint32_t key[8]) {
    for (int i = 0; i < 7; i++)
        base[REG_KEY0/4 + i] = key[i]; // KEY[0..6]
    base[REG_KEY7/4] = key[7];          // Triggers key expansion
    while (!(base[REG_STATUS/4] & STATUS_KEY_READY)); // Poll
}
```

6.3 Single-Block CBC Encryption

4. **Step 1:** Ensure `KEY_READY` = 1.
5. **Step 2:** Write `IV[0..3]` with the 128-bit initialization vector.
6. **Step 3:** Write `CTRL` with `MODE=001` (CBC), `ENC_DEC=1` (encrypt), `INIT_IV=1`.
7. **Step 4:** Write `IRQ_EN` = 0x01 to enable `DOUT_VLD` interrupt.
8. **Step 5:** Write `DIN_CTRL` = 0x01 (`DIN_LAST` = 1, this is the last and only block).
9. **Step 6:** Write `DIN[0]` through `DIN[2]` with plaintext words 0–2.
10. **Step 7:** Write `DIN[3]` with plaintext word 3. This write simultaneously triggers encryption.
11. **Step 8:** Service interrupt (or poll `STATUS[1]` `DONE`). Read `DOUT[0..3]` for ciphertext.
12. **Step 9:** Write `IRQ_STAT` = 0x01 to clear interrupt.

6.4 Multi-Block CBC Streaming

For multi-block messages, the back-to-back streaming feature eliminates the need to re-load the IV or re-set `INIT_IV` between blocks. The internal chaining register is automatically updated after each block.

13. **Step 1:** Load key and IV as in Section 6.2 and Step 2 of 6.3.
14. **Step 2:** Write `CTRL` with `MODE=001`, `ENC_DEC=1`, `INIT_IV=1`.
15. **Step 3:** Write `IRQ_EN` = 0x01.
16. **Steps 4–N (for each block except the last):** Write `DIN_CTRL` = 0x00 (`DIN_LAST` cleared). Write `DIN[0..3]`. Wait for interrupt. Read `DOUT[0..3]`. Write `IRQ_STAT` = 0x01. Do NOT rewrite `CTRL` or IV.

17. **Step N+1 (last block):** Write DIN_CTRL = 0x01 (DIN_LAST = 1). Write DIN[0..3]. Wait for interrupt. Read DOUT[0..3]. Write IRQ_STAT = 0x01.

NOTE

After clearing IRQ_STAT[0] (DOUT_VLD), the core latches the cleared state and begins accepting the next DIN block. Writing the next DIN[3] before IRQ_STAT is cleared will cause the data to be ignored and the write to return SLVERR.

6.5 GCM Authenticated Encryption

GCM operation proceeds in three phases: AAD processing, ciphertext processing, and tag finalization.

Phase 1 — AAD Processing

18. **Step 1:** Write CTRL with MODE=101 (GCM), ENC_DEC=1, INIT_IV=1.
19. **Step 2:** Write IV[0..3] with the 96-bit nonce (IV[0..2] = nonce, IV[3] = 0x00000001 per GCM spec).
20. **Step 3:** Write GCM_LEN_A and GCM_LEN_A_HI with the total AAD length in bits.
21. **Step 4:** Write GCM_LEN_C and GCM_LEN_C_HI with the total ciphertext length in bits.
22. **Step 5:** For each AAD block except the last: write DIN_CTRL = 0x00. Write AAD[0..3]. GHASH update is triggered by the AAD[3] write.
23. **Step 6:** For the last AAD block: write DIN_CTRL = 0x02 (AAD_LAST = 1). Write AAD[0..3].

Phase 2 — Ciphertext Processing

24. **Step 7:** For each plaintext block except the last: write DIN_CTRL = 0x00. Write DIN[0..3]. Wait for DOUT_VLD interrupt. Read DOUT. Clear IRQ_STAT.
25. **Step 8:** For the last plaintext block: write DIN_CTRL = 0x01 (DIN_LAST = 1). Write DIN[0..3].

Phase 3 — Tag Finalization

26. **Step 9:** Wait for both DOUT_VLD (IRQ_STAT[0]) and TAG_VLD (IRQ_STAT[1]) interrupts.
27. **Step 10:** Read DOUT[0..3] for the last ciphertext block.
28. **Step 11:** Read TAG[0..3] for the 128-bit authentication tag.
29. **Step 12:** Write IRQ_STAT = 0x03 to clear both interrupt flags.

```
// GCM encryption skeleton (C pseudocode)
void aes_gcm_encrypt(aes_ctx_t *ctx, const uint8_t *aad, size_t aad_len,
                    const uint8_t *pt, size_t pt_len,
                    uint8_t *ct, uint8_t tag[16]) {
    aes_write32(ctx, REG_CTRL, CTRL_MODE_GCM | CTRL_ENC | CTRL_INIT_IV);
    aes_load_iv(ctx, ctx->nonce);
    aes_write32(ctx, REG_GCM_LEN_A, (uint32_t)(aad_len * 8));
    aes_write32(ctx, REG_GCM_LEN_A_HI, (uint32_t)((aad_len * 8) >> 32));
    aes_write32(ctx, REG_GCM_LEN_C, (uint32_t)(pt_len * 8));
    aes_write32(ctx, REG_GCM_LEN_C_HI, (uint32_t)((pt_len * 8) >> 32));
    aes_process_aad(ctx, aad, aad_len);
    aes_process_blocks(ctx, pt, pt_len, ct);
    aes_read_tag(ctx, tag);
}
```

7. Timing Specifications

7.1 Clock and Reset Requirements

Parameter	Min	Max	Unit
Clock Frequency (f_clk)	10	400	MHz
Clock Duty Cycle	45	55	%
Reset Pulse Width (aresetn low)	2	—	clock cycles
Key Expansion Latency	60	62	clock cycles
Block Processing Latency (ECB/CBC/CTR)	14	14	clock cycles
GHASH Update Latency (per AAD block)	4	4	clock cycles
Tag Finalization Latency (GCM)	20	22	clock cycles
AXI Write-to-BVALID Latency	1	2	clock cycles
AXI Read-to-RVALID Latency	1	2	clock cycles

7.2 Throughput Analysis

Peak throughput (ignoring AXI bus overhead) is calculated as:

$$\text{Throughput} = (128 \text{ bits} / \text{block}) \times (f_{\text{clk}} / 14 \text{ cycles/block})$$

At 400 MHz: Throughput $\approx$ **3.66 Gbps**. At 200 MHz: Throughput $\approx$ 1.83 Gbps. In back-to-back streaming with optimal bus timing (4 write cycles for DIN + 1 cycle IRQ latency), effective throughput with bus overhead is approximately **82% of peak** at 400 MHz.

7.3 Timing Diagram — Single Block Operation

The following describes the cycle-accurate timing for a single CBC encryption block (abbreviated for readability):

Cycle	Signal	Event
0	i_aes_awaddr=0x040 i_aes_awvalid=1 i_aes_wdata=PLAIN[31:0] i_aes_wvalid=1	DIN[0] write address
1	o_aes_awready=1 o_aes_wready=1 o_aes_bvalid=1 BRESP=00	AIC accepts address AIC accepts data Write ACK
1-3	... (DIN[1..2] writes, same pattern)	
4	i_aes_awaddr=0x04C i_aes_wdata=PLAIN[127:96]	DIN[3] write — TRIGGERS AES
5	STATUS[BUSY]=1	AES datapath started
18	STATUS[BUSY]=0 STATUS[DONE]=1 IRQ_STAT[DOUT_VLD]=1	14 rounds complete
	o_aes_irq=1	IRQ fired
19	CPU reads DOUT[0..3]	Output consumed
19	i_aes_wdata=0x1	Write IRQ_STAT[0]=1 (W1C)
20	IRQ_STAT[DOUT_VLD]=0 o_aes_irq=0	IRQ cleared

8. Electrical and Physical Characteristics

8.1 Technology and Process

Parameter	Value / Description
Target Process	TSMC 28nm HPC+ (High Performance Compact Plus)
Voltage (VDD)	0.9 V nominal (0.81 V min, 0.99 V max)
Core Area (estimated)	~0.04 mm ² (40 kGE equivalent gate count)
Dynamic Power (400 MHz, 1.0 data toggle)	~28 mW (estimated; PPA to be updated post-PnR)
Leakage Power (typical corner)	~1.2 mW
Critical Path (worst-case SS/0.81V/125°C)	ADP MixColumns → AddRoundKey (target: 2.5 ns)
Scan Coverage	≥98% (excluding ROM cells)

8.2 Clock Domain Structure

The IP is a single-clock-domain design. All flip-flops and memory arrays are clocked by `i_aes_aclk`. There are no asynchronous clock-domain crossings within this IP.

NOTE

If integrated into an SoC where the AXI fabric clock and the crypto engine clock are separate, the integrator must instantiate an appropriate CDC synchronization module at the IP boundary and provide a bridge. This is outside the scope of this IP.

8.3 Memory Content (Round Key RAM)

The round-key storage is implemented as a synchronous single-port SRAM with the following parameters:

Parameter	Value
Depth	60 words
Width	32 bits
Total Size	240 bytes (1920 bits)
Read Latency	1 clock cycle (synchronous)
Write Latency	1 clock cycle (synchronous)
Retention on Reset	Contents cleared on <code>i_aes_aresetn</code> assertion.

9. SoC Integration Guidelines

9.1 Instantiation Template (SystemVerilog)

```
// AES-256 IP Core - SV instantiation template
asi_aes256_top #(
    .C_S_AXI_DATA_WIDTH (32),
    .C_S_AXI_ADDR_WIDTH (8)
) u_aes256 (
    // Global
    .i_aes_aclk      (soc_clk      ),
    .i_aes_aresetn   (soc_resetn   ),
    // AXI4-Lite Write Address Channel
    .i_aes_awvalid   (axi_awvalid   ),
    .o_aes_awready   (axi_awready   ),
    .i_aes_awaddr    (axi_awaddr[7:0]),
    .i_aes_awprot     (axi_awprot    ),
    // AXI4-Lite Write Data Channel
    .i_aes_wvalid    (axi_wvalid    ),
    .o_aes_wready    (axi_wready    ),
    .i_aes_wdata     (axi_wdata     ),
    .i_aes_wstrb     (axi_wstrb     ),
    // AXI4-Lite Write Response Channel
    .o_aes_bvalid    (axi_bvalid    ),
    .i_aes_bready    (axi_bready    ),
    .o_aes_bresp     (axi_bresp     ),
    // AXI4-Lite Read Address Channel
    .i_aes_arvalid   (axi_arvalid   ),
    .o_aes_arready   (axi_arready   ),
    .i_aes_araddr    (axi_araddr[7:0]),
    .i_aes_arprot     (axi_arprot    ),
    // AXI4-Lite Read Data Channel
    .o_aes_rvalid    (axi_rvalid    ),
    .i_aes_rready    (axi_rready    ),
    .o_aes_rdata     (axi_rdata     ),
    .o_aes_rresp     (axi_rresp     ),
    // Interrupt
    .o_aes_irq       (aes_irq       )
);
```

9.2 Address Mapping Example

In a typical ARM Cortex-A55 SoC, the AES IP base address might be assigned as follows:

```
// Device Tree Snippet (DTS)
aes_crypto: aes@40010000 {
    compatible = "asi,aes256-1.0";
    reg = <0x0 0x40010000 0x0 0x100>;
    interrupts = <GIC_SPI 42 IRQ_TYPE_LEVEL_HIGH>;
    interrupt-parent = <&gic>;
    clocks = <&periph_clk>;
    clock-names = "apb_pclk";
    status = "okay";
};
```

9.3 Integration Checklist

- Verify that the AXI4-Lite master does not issue outstanding transactions to this slave.
- Ensure i_aes_aresetn is held low for a minimum of 2 clock cycles during power-on reset.
- Connect o_aes_irq to an interrupt controller input configured for level-high sensitivity.
- Assign a unique interrupt number and configure the CPU's GIC or PLIC accordingly.

- Apply a full wstrb (4'hF) for all register writes; partial strobes will be rejected with SLVERR.
- Do not assert wvalid before avalid on any AXI4-Lite transaction.
- If DMA is used to feed DIN data, ensure DMA bursts terminate at DIN[3] (offset 0x04C) to trigger processing.
- Confirm KEY registers are written in order KEY[0] through KEY[7] within a single key-loading sequence.
- For GCM, always write GCM_LEN registers before starting AAD processing.
- Validate against all NIST CAVP Known-Answer Test vectors included in the IP delivery package.

10. Verification Plan Overview

The AES-256 IP Core is verified using a UVM-compliant verification environment. The following table summarizes the key verification components and coverage targets.

Component	Method	Coverage Target
FIPS-197 Compliance	Directed tests using all NIST CAVP vectors.	100% vector pass rate.
Register Map	UVM RAL model — read/write/reset checks.	100% register field coverage.
All Six Modes	Directed + constrained-random block sequences.	≥10,000 random blocks per mode.
GCM Tag Verification	Reference model comparison (C OpenSSL golden model).	100% pass on NIST GCM test vectors.
AXI4-Lite Protocol	Protocol checker assertions (SVA).	0 protocol violations.
Back-to-Back Streaming	Burst sequences of 1–1024 blocks.	Data integrity verified on all burst sizes.
Reset Behavior	Mid-operation reset injection.	Core recovers cleanly in ≤3 cycles.
Code Coverage	Line, branch, toggle, FSM coverage.	≥95% code coverage.

Appendix A — Register Quick Reference Card

This appendix provides a condensed one-page register reference for firmware developers.

Offset	Name	RW	Reset / Key Bits
0x000	CTRL	R/W	0x00 [0]=START(W1S), [1]=ENC_DEC, [4:2]=MODE, [5]=INIT_IV
0x004	STATUS	R	0x00 [0]=BUSY, [1]=DONE, [2]=KEY_READY, [3]=AUTH_FAIL
0x008	IRQ_EN	R/W	0x00 [0]=DOUT_VLD_EN, [1]=TAG_VLD_EN
0x00C	IRQ_STAT	R/W1C	0x00 [0]=DOUT_VLD, [1]=TAG_VLD — Write 1 to clear
0x010	KEY[0]	W	— Key bits [31:0]
0x02C	KEY[7]	W	— Key bits [255:224] — triggers expansion
0x030	IV[0]	W	— IV bits [31:0]
0x03C	IV[3]	W	— IV bits [127:96]
0x040	DIN[0]	W	— Input data bits [31:0]
0x04C	DIN[3]	W	— Input data bits [127:96] — triggers AES
0x050	DOUT[0]	R	— Output bits [31:0]
0x05C	DOUT[3]	R	— Output bits [127:96]
0x060	AAD[0]	W	— AAD bits [31:0] (GCM only)
0x06C	AAD[3]	W	— AAD bits [127:96] — triggers GHASH
0x070	TAG[0]	R	— Auth Tag bits [31:0] (GCM only)
0x07C	TAG[3]	R	— Auth Tag bits [127:96]
0x080	DIN_CTRL	R/W	0x00 [0]=DIN_LAST, [1]=AAD_LAST
0x084	GCM_LEN_A	W	— AAD length in bits [31:0]
0x088	GCM_LEN_A_HI	W	— AAD length in bits [63:32]
0x08C	GCM_LEN_C	W	— Cipher length in bits [31:0]
0x090	GCM_LEN_C_HI	W	— Cipher length in bits [63:32]
0x0FC	VERSION	R	0x00010000 [31:16]=Major, [15:8]=Minor, [7:0]=Patch

Appendix B — Known-Answer Test Vectors (NIST Subset)

The following vectors are excerpted from the NIST CAVP AES-256 test suite. All values are hexadecimal. These vectors must pass on the RTL before tape-out sign-off.

B.1 AES-256 ECB Encrypt

Field	Value (hex)
Key	000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f
Plaintext	00112233445566778899aabbccddeeff
Ciphertext	8ea2b7ca516745bfeafc49904b496089

B.2 AES-256 CBC Encrypt (NIST SP 800-38A F.2.6)

Field	Value (hex)
Key	603deb1015ca71be2b73aef0857d77811f352c073b6108d72d9810a30914dff4
IV	000102030405060708090a0b0c0d0e0f
Block 1 PT	6bc1bee22e409f96e93d7e117393172a
Block 1 CT	f58c4c04d6e5f1ba779eabfb5f7bfbd6
Block 2 PT	ae2d8a571e03ac9c9eb76fac45af8e51
Block 2 CT	9cfc4e967edb808d679f777bc6702c7d

B.3 AES-256 GCM Encrypt (NIST SP 800-38D Test Case 15)

Field	Value (hex)
Key	feffe9928665731c6d6a8f9467308308feffe9928665731c6d6a8f9467308308
IV (96-bit)	cafebabefacedbaddecaf888
AAD	feedfacedeadbeeffeedfacedeadbeefabaddad2
Plaintext	d9313225f88406e5a55909c5aff5269a86a7a9531534f7da2e4c303d8a318a72...
Ciphertext	522dc1f099567d07f47f37a32a84427d643a8cdcbfe5c0c97598a2bd2555d1aa...
Tag (128-bit)	76fc6ece0f4e1768cddf8853bb2d551b

Appendix C — Glossary

Active-Low Reset: A reset signal that is asserted when at logic 0. The device is in reset while the signal is low.

AMBA: Advanced Microcontroller Bus Architecture — ARM's open standard interconnect specification for SoC design.

AXI4-Lite: A simplified subset of the AXI4 protocol optimized for low-throughput, low-latency register-mapped interfaces.

Back-pressure: A flow-control mechanism where a downstream component signals to an upstream component to pause data transmission until the downstream is ready.

CAVP: Cryptographic Algorithm Validation Program — NIST's program for validating cryptographic implementations.

Chaining Value: The intermediate state (e.g., ciphertext block in CBC, counter in CTR) maintained between consecutive block operations in a multi-block message.

GCM: Galois/Counter Mode — an authenticated encryption mode that combines CTR mode with GHASH to provide both confidentiality and data authenticity.

GHASH: The universal hash function used in GCM, defined over $GF(2^{128})$.

Iterative Datapath: A hardware architecture where the same combinational logic is reused for multiple sequential operations (rounds), trading throughput for area.

KAT: Known-Answer Test — a test that validates an implementation using pre-defined input/output pairs derived from the standard.

Nonce: Number Used Once — a unique value supplied to an encryption algorithm per-message to ensure that the same plaintext always produces different ciphertext.

S-Box: Substitution Box — the non-linear byte substitution table used in the AES SubBytes and InvSubBytes transformations.

W1C: Write-1-to-Clear — a register field access type where writing a 1 to a bit clears it to 0; writing 0 has no effect.

W1S: Write-1-to-Set — a register field access type where writing a 1 sets the bit; writing 0 has no effect.